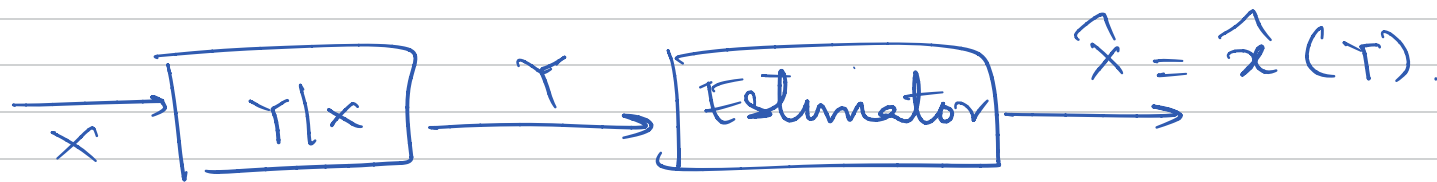


Lec 39

① MMSE estimator

$$\hat{x}_{\text{MMSE}}(y) = E[X | Y=y]$$



MMSE estimator minimizes the MSE
(mean square error)

$$\begin{aligned} \text{MSE} &= E[(x - \hat{x})^2] \\ \hat{x}_{\text{MMSE}}^{(i)} &= \arg \min_{\hat{x}(\cdot)} E[(x - \hat{x}(Y))^2] \end{aligned}$$

② Best Constant estimator

$$\hat{x}_{BCE}(y) = E[X] \quad (\text{not dependent on value of } y).$$

$$\begin{aligned} \text{MSE seen by BCE estimator} &= E[(X - \hat{x}_{BCE}(r))^2] \\ &= E[(X - E[X])^2] \\ &= \text{Var}(X). \end{aligned}$$

Example: $X \sim \text{Uniform}([4, 10])$

$W \sim \text{Uniform}([-1, 1])$

assume W is
independent of X .
 $W \perp X$

$$Y = X + W$$

① $\hat{x}_{\text{BCE}}(y) = E[X] = 7.$

$$\text{MSE}_{\text{BCE}} = E[(X - \hat{x}_{\text{BCE}}(Y))^2] = \text{Var}(X) = 3.$$

② $\hat{x}_{\text{MMSE}}(y) = E[X | Y=y].$

• find $f_{X|Y}(x|y).$

• find $E[X | Y=y] = E[Y - W | Y=y].$
 $= y - E[W | Y=y]$

$$(x, w) \iff (x, y).$$

$$f_{x, y}(x, y) = \frac{f_{x, w}(x, w)}{J_{x, w}(x, y)}$$

$$J_{x, w}(x, y) = \begin{vmatrix} \frac{\partial x}{\partial x} & \frac{\partial y}{\partial x} \\ \frac{\partial x}{\partial w} & \frac{\partial y}{\partial w} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1.$$

$$\Rightarrow f_{x, y}(x, y) = f_{x, w}(x, y-x)$$

$$= f_x(x) f_w(y-x).$$

$$= \frac{1}{6} \times \frac{1}{2} \quad \begin{matrix} x \in [4, 10] \\ \& y-x \in [-1, 1] \end{matrix}.$$

$$\Rightarrow f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}.$$

$$= \frac{\frac{1}{2} \times \frac{1}{6}}{\frac{1}{6}} \quad \begin{array}{l} x \in [4, 10] \\ y \in [x-1, x+1] \end{array}$$

$$= \frac{1}{2}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx.$$

$\Uparrow$ $y \in [x-1, x+1]$
 $x \in [4, 10]$
 $x \in [y-1, y+1]$
 $\& x \in [4, 10]$

$$= \int_{\max(4, y-1)}^{\min(y+1, 10)} \frac{1}{2} \times \frac{1}{6} dx.$$

$$\frac{1}{12} \left[\min(y+1, 10) - \max(4, y-1) \right] = \int_{y-1}^{y+1} \frac{1}{2} \times \frac{1}{6} dx = \frac{1}{6} \quad 5 \leq y \leq 9.$$

Sanity check

$$\int_{-\infty}^{\infty} f_Y(y) dy = 1$$

$$\int_{y-1}^{10} \frac{1}{2} \times \frac{1}{6} dx = \frac{11-y}{12} \quad 11 \geq y > 9$$

$$\int_4^{y+1} \frac{1}{2} \times \frac{1}{6} dx = \frac{y-3}{12} \quad 3 \leq y < 5$$

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f(y)}$$

$$y-1 \leq x \leq y+1$$

$$5 \leq y \leq 9$$

$$9 \leq y \leq 11$$

$$y-1 \leq x \leq 10$$

$$3 \leq y < 5$$

$$4 \leq x \leq y+1$$

$f_{X,Y}(x,y) = \begin{cases} \text{Uniform}[y-1, y+1] & \text{for } 5 \leq y \leq 9 \\ \text{Uniform}[y-1, 10] & \text{for } 9 \leq y \leq 11 \\ \text{Uniform}[4, y+1] & \text{for } 3 \leq y < 5 \end{cases}$

$$\frac{\frac{1}{2} \times \frac{1}{6}}{\frac{1}{6}} = \frac{1}{2}$$

$$\frac{\frac{1}{2} \times \frac{1}{6}}{\frac{11-y}{12}} = \frac{1}{11-y}$$

$$\frac{\frac{1}{2} \times \frac{1}{6}}{\frac{y-3}{12}} = \frac{1}{y-3}$$

$$\hat{x}_{\text{MMSE}}(y) = E[x | Y=y] = \begin{cases} y & 5 \leq y \leq 9 \\ \frac{9+y}{2} & 1 \leq y \leq 11 \\ \frac{y+5}{2} & 3 \leq y \leq 5 \end{cases}$$

© Suppose $Y = X + W$.

$$X \sim N(\mu_x, \sigma_x^2)$$

$$W \sim N(0, \sigma^2). \quad X \perp W.$$

Find MMSE estimate of x from Y .

$$f_{Y|X}(y|x) = f_W(y-x) = \frac{e^{-\left(\frac{y-x}{\sigma}\right)^2 \frac{1}{2}}}{\sqrt{2\pi}\sigma}.$$

$$\Rightarrow Y|X=x \sim N(x, \sigma^2).$$

$$Y \sim N(\mu_x, \sigma_x^2 + \sigma^2)$$

→ Sum of independent Gaussian is Gaussian R.V.

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x) f_X(x)}{f_Y(y)}$$

$f(x) = ax \rightarrow$ linear fn
 $f(x) = ax+b \rightarrow$ affine fn

$$\propto e^{-\frac{1}{2} \left(\frac{1}{\sigma^2} + \frac{1}{\sigma_x^2} \right) \left(x - \frac{\frac{y}{\sigma^2} + \frac{\mu_x}{\sigma_x^2}}{\frac{1}{\sigma^2} + \frac{1}{\sigma_x^2}} \right)^2}$$

$$\propto e^{-\frac{(y-x)^2}{2\sigma^2}} e^{-\frac{1}{2} \left(\frac{x-\mu_x}{\sigma_x} \right)^2}$$

$$\propto e^{-\frac{1}{2} \left[\frac{1}{\sigma^2} + \frac{1}{\sigma_x^2} \right] x^2 - 2x \left[\frac{y}{\sigma^2} + \frac{\mu_x}{\sigma_x^2} \right]}$$

$$\propto e^{-\frac{1}{2} \left(\frac{1}{\sigma^2} + \frac{1}{\sigma_x^2} \right) \left[x^2 - 2x \left(\frac{\frac{y}{\sigma^2} + \frac{\mu_x}{\sigma_x^2}}{\left(\frac{1}{\sigma^2} + \frac{1}{\sigma_x^2} \right)} \right) \right]}$$

$$\hat{x}_{\text{MMSE}}(y) = E[X|Y=y] = \frac{\left[\frac{y}{\sigma^2} + \frac{\mu_x}{\sigma_x^2} \right]}{\left(\frac{1}{\sigma^2} + \frac{1}{\sigma_x^2} \right)}$$

Linear MMSE estimator

$$\hat{x}_{\text{LMMSE}}(y) = a^* y + b^* \text{ st.}$$

$$(a^*, b^*) = \arg \min_{(a, b)} E[(x - (aY + b))^2].$$

$$E[(x - aY - b)^2] \geq E[(x - aY - (E[x - aY]))^2]$$

$$= E[(\bar{x} - a\bar{Y})^2].$$

$$\text{where } \bar{x} = x - E[x]$$

$$\bar{Y} = Y - E[Y].$$

$$b^* = E[x - a^* Y]$$

$$a^* = \arg \min_a E[\underbrace{(\bar{x} - a\bar{Y})^2}_{f(a)}]$$

$$f(a) = E[\bar{X}^2 + a^2 \bar{Y}^2 - 2a \bar{X} \bar{Y}]$$

$$= E[\bar{X}^2] + a^2 E[\bar{Y}^2] - 2a E[\bar{X} \bar{Y}]$$

$$f'(a) = 2a E[\bar{Y}^2] - 2 E[\bar{X} \bar{Y}]$$

$$f''(a) = 2 E[\bar{Y}^2] \geq 0$$

$$\Rightarrow a^* = \frac{\text{Cov}(X, Y)}{\text{Var}(Y)}$$

MSE seen by linear estimators.

minimizes the

$$\begin{aligned} \hat{x}_{\text{LMMSE}}(y) &= a^* Y + b^* = a^* Y + E[X - a^* Y] \\ &= a^* (Y - E[Y]) + E[X] = \frac{\text{Cov}(X, Y)}{\text{Var}(Y)} (Y - E[Y]) + E[X] \end{aligned}$$

Example:

(a) $X \sim \text{Uniform}[4, 10]$.

$$W \sim \text{Uniform}[-1, 1]$$

$$Y = X + W$$

$$\hat{x}_{\text{LMMSE}}(y) = \frac{3}{\left(\frac{10}{3}\right)} (y - 7) + 7.$$

$$= \frac{9}{10} (y - 7) + 7.$$

$$E[X] = 7, \quad E[W] = 0.$$

$$E[Y] = 7.$$

$$W \perp X.$$

$$\begin{aligned} \text{Cov}(X, Y) &= E[XY] - E[X]E[Y] \\ &= E[X^2 + XW] - 7^2 \\ &= E[X^2] - 7^2 \\ &= 3 + 7^2 - 7^2 \end{aligned}$$

$$\begin{aligned} \text{Var}(Y) &= \text{Var}(X) + \text{Var}(W) \\ &= 3 + \frac{1}{3} = \frac{10}{3}. \end{aligned}$$

$$\begin{aligned} \int_{-1}^1 \frac{w^2}{2} dw \\ = \frac{w^3}{6} \Big|_{-1}^1 = \frac{2}{6} = \frac{1}{3}. \end{aligned}$$

⑥

$$Y = X + W$$

$$X \sim N(\mu_x, \sigma_x^2)$$

$$W \sim N(0, \sigma^2)$$

$$E[Y] = E[X] + E[W]$$

$$E[X] = \mu_x$$

$$\begin{aligned} \text{Cov}(X, Y) &= E[XY] - \mu_x^2 \\ &= E[X^2 + XW] - \mu_x^2 = \text{Var}(X) = \sigma_x^2 \end{aligned}$$

$$\text{Var}(Y) = \sigma_x^2 + \sigma^2$$

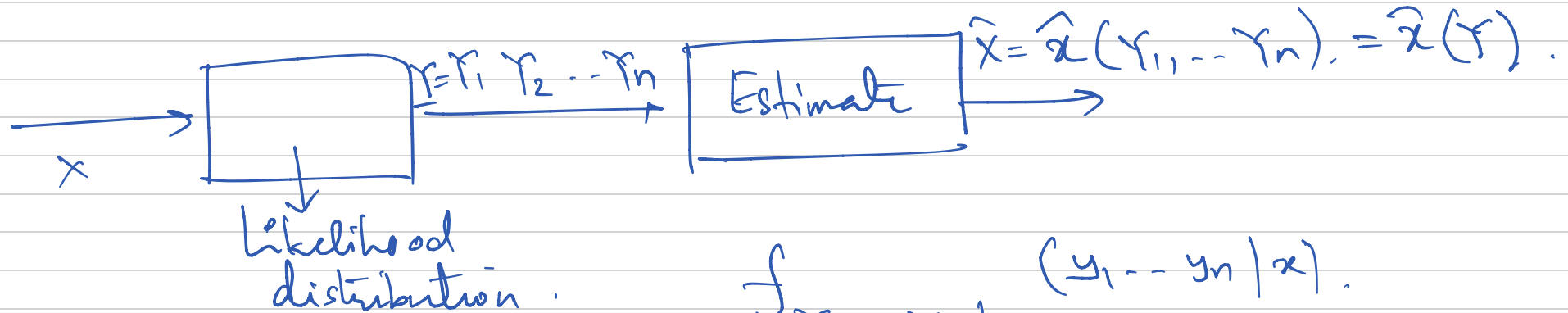
$$\hat{x}_{\text{LMMSE}}(y) = \frac{\sigma_x^2}{\sigma_x^2 + \sigma^2} (y - \mu_x) + \mu_x$$

$$= \left(\frac{\sigma_x^2}{\sigma_x^2 + \sigma^2} y + \mu_x \left(\frac{\sigma^2}{\sigma_x^2 + \sigma^2} \right) \right)$$

$$= \left(\frac{y}{\sigma^2} + \frac{\mu n}{\sigma_n^2} \right) \frac{\sigma^2 \sigma_n^2}{\sigma_n^2 + \sigma^2}$$

$$= \hat{x}_{\text{MMSE}}(y).$$

Multiple measurements



$$f_{y_1, \dots, y_n | x}(y_1, \dots, y_n | x)$$

and $f_x(x)$.

If the joint distribution is not specified then it is mostly assumed that

$$f_{y_1, \dots, y_n | x}(y_1, \dots, y_n | x) = f_{y_1}(y_1 | x) f_{y_2}(y_2 | x)$$

$Y_1, \dots, Y_n \mid X=x$ are assumed to be iid

$\dots f_{Y_n}(y_n|x)$
are assumed to be the same.

Programming assignment.

$$Y_i = h_i^0 X + W_i$$

$$f_{Y_i|X}(y_i|x) = f_{W_i}(y_i - h_i^0 x)$$

(Exercise).

(a) MMSE estimation

$$\arg \min_{\hat{x}(\cdot)} E[(X - \hat{x}(Y_1, \dots, Y_n))^2] = \hat{x}_{\text{MMSE}}(\cdot)$$

$$\hat{x}_{\text{MMSE}}(y_1, \dots, y_n) = E[X \mid Y_1 = y_1, Y_2 = y_2, \dots, Y_n = y_n]$$

$$E[(X - \hat{x}(Y_1, \dots, Y_n))^2]$$

$$= \int f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) \left[E \left[\left(x - \hat{a}(y_1, \dots, y_n) \right)^2 \mid Y_1 = y_1, \dots, Y_n = y_n \right] \right] dy_1, \dots, dy_n.$$

$$\geq \int f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) E \left[\left(x - E[x \mid Y_1 = y_1, \dots, Y_n = y_n] \right)^2 \mid Y_1 = y_1, \dots, Y_n = y_n \right] dy_1, \dots, dy_n.$$

⑥ LMMSE estimator.

$$\hat{x}_{\text{LMMSE}}(y_1, \dots, y_n) = \sum_{i=1}^n a_i^* r_i + b^*$$

$$(a_1^* \dots a_n^* b^*) = \arg \min_{(a_1, \dots, a_n, b)} E \left[(x - \sum a_i r_i - b)^2 \right]$$

$$E \left[(x - \sum a_i r_i - b)^2 \right] \geq E \left[(\bar{x} - \sum a_i \bar{r}_i)^2 \right]$$
$$\downarrow$$
$$b = E[x - \sum a_i r_i].$$

$$(a_1^*, \dots, a_n^*) = \arg \min_{(a_1, \dots, a_n)} E \left[\left(\bar{x} - \sum_{i=1}^n a_i \bar{r}_i \right)^2 \right]$$

$$\frac{\partial f}{\partial a_j} = E \left[\frac{\partial}{\partial a_i} \left(\bar{x} - \sum_{i=1}^n a_i \bar{r}_i \right)^2 \right]$$

$$= E \left[-2 \left(\bar{x} - \sum_{i=1}^n a_i \bar{y}_i \right) \bar{y}_j \right].$$

$$= 0.$$

Orthogonality Condition: Let $\hat{\bar{x}}_{\text{LMMSE}}$ be the LMMSE estimator.

$$E \left[\left(\bar{x} - \hat{\bar{x}}_{\text{LMMSE}} \right) \bar{y}_j \right] = 0$$

for all $j=1, 2, \dots, n$.



$$E \left[\bar{x} \bar{y}_j \right] = \sum_{i=1}^n a_i E \left[\bar{y}_i \bar{y}_j \right]$$

$$\text{Cov}(\bar{x}, \bar{y}_j) = \sum_{i=1}^n a_i \text{Cov}(\bar{y}_i, \bar{y}_j) = 0$$

As you vary j there are n equations and
 n unknowns.

$$\begin{bmatrix} \text{Cov}(Y_1, Y_1) & \text{Cov}(Y_2, Y_1) & \dots & \text{Cov}(Y_n, Y_1) \\ \text{Cov}(Y_1, Y_2) & \text{Cov}(Y_2, Y_2) & \dots & \text{Cov}(Y_n, Y_2) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(Y_1, Y_n) & \dots & \dots & \text{Cov}(Y_n, Y_n) \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} \text{Cov}(X, Y_1) \\ \vdots \\ \text{Cov}(X, Y_n) \end{bmatrix}$$

MSE Analysis.

① BCE

$$\begin{aligned} \text{MSE}_{\text{BCE}} &= E \left[(X - \hat{X}_{\text{BCE}})^2 \right] \\ &= E \left[(X - E[X])^2 \right] = \text{Var}(X). \end{aligned}$$

② MMSE estimator $\hat{X}_{\text{MMSE}} = \hat{x}_{\text{MMSE}}(Y) = E[X|Y]$.

$$\begin{aligned} \text{MSE}_{\text{MMSE}} &= E \left[(X - \hat{X}_{\text{MMSE}})^2 \right] \\ &= E \left[(X - E[X|Y])^2 \right] \\ &= E \left[\underbrace{E[(X - E[X|Y])^2 | Y]}_{\text{Var}(X|Y)} \right] \\ &= E \left[\text{Var}(X|Y) \right] \end{aligned}$$

From total law of variance we know

$$\underbrace{\text{Var}(X)}_{\text{MSE}_{\text{BCE}}} = \underbrace{E[\text{Var}(X|Y)]}_{\text{MSE}_{\text{MMSE}}} + \underbrace{\text{Var}(E[X|Y])}_{\geq 0}.$$

$$\Rightarrow \text{MSE}_{\text{MMSE}} \leq \text{MSE}_{\text{BCE}}$$

$$\begin{aligned}\text{MSE}_{\text{MMSE}} &= E[\text{Var}(X|Y)] \\ &= \text{Var}(X) - \text{Var}(E[X|Y]) \\ &= \text{Var}(X) - \text{Var}(\hat{X}_{\text{MMSE}}).\end{aligned}$$

Suppose X is independent of Y .

$$\hat{X}_{\text{MMSE}} = E[X|Y] = E[X] = \hat{X}_{\text{BCE}}$$

$$\text{Var}(E[X|Y]) = \text{Var}(E[X]) = 0.$$

$$MSE_{MMSE} = \text{Var}(x).$$

def $\tilde{x}_{MMSE} = x - \hat{x}_{MMSE}$

error

$$E[\tilde{x}_{MMSE}] = E[x - \hat{x}_{MMSE}]$$

$$= E[x - E[x|Y]]$$

$$= E[x] - E[E[x|Y]]$$

total law
of expectation $= E[x] - E[x]$
 $= 0.$

$$E[\tilde{x}_{MMSE} | Y] = E[x - \hat{x}_{MMSE} | Y]$$

$$= E[x - \underbrace{E[x|Y]} | Y]$$

$$\text{If } E[\tilde{x}] = 0$$

then $\hat{x}$ is
referred to
as unbiased
estimator.

$$\begin{aligned}
&= E[x - f(r) | r] \\
&= E[x | r] - E[f(r) | r] \\
&= E[x | r] - f(r). \\
&= E[x | r] - E[x | r] = 0.
\end{aligned}$$

③ LMMSE estimate.

$$\hat{x}_{\text{LMMSE}} = \hat{x}_{\text{LMMSE}}(r) = \frac{\text{Cov}(x, r)}{\text{Var}(r)} (r - E[r]) + E[x] \quad // a r + b.$$

$$a = \frac{\text{Cov}(x, r)}{\text{Var}(r)}.$$

$$\begin{aligned}
\text{MSE}_{\text{LMMSE}} &= E[(x - \hat{x}_{\text{LMMSE}})^2] \\
&= E[(\bar{x} - a \bar{r})^2].
\end{aligned}$$

$$= E[\bar{X}^2 + a^2 \bar{Y}^2 - 2a \bar{X} \bar{Y}].$$

$$= \text{Var}(X) + a^2 \text{Var}(Y) - 2a \text{Cov}(X, Y).$$

$$= \text{Var}(X) + \frac{\text{Cov}(X, Y)^2}{\text{Var}(Y)} - \frac{2a \text{Cov}(X, Y)}{\text{Var}(Y)}.$$

$$= \text{Var}(X) - \frac{\text{Cov}(X, Y)^2}{\text{Var}(Y)} \leq \text{Var}(X)$$

To show $\text{MSE}_{\text{BCE}} \geq \text{MSE}_{\text{LMMSE}} \geq \text{MSE}_{\text{MMSE}}$
↓
to be proved.

$$\text{MSE}_{\text{MMSE}} = \text{Var}(X) - \text{Var}(E[X|Y]).$$

Enough to show that $\text{Var}(E[X|Y]) \geq \frac{\text{Cov}(X, Y)^2}{\text{Var}(Y)}.$

ie., to show $\text{Var}(\hat{x}_{\text{MMSE}}) \text{Var}(Y) \geq \text{Cov}(x, Y)^2$

We know that $\text{Cov}(\hat{x}_{\text{MMSE}}, Y)^2 \leq \text{Var}(\hat{x}_{\text{MMSE}}) \text{Var}(Y)$
as correlation $\in [-1, 1] \rightarrow \textcircled{1}$

$$\text{ie., } \rho^2(\hat{x}_{\text{MMSE}}, Y) \leq 1.$$

We'll show that $\text{Cov}(x, Y) = \text{Cov}(\hat{x}_{\text{MMSE}}, Y)$. $\rightarrow \textcircled{2}$

From $\textcircled{1}$ & $\textcircled{2}$ it follows that

$$\text{Cov}(x, Y)^2 \leq \text{Var}(\hat{x}_{\text{MMSE}}) \text{Var}(Y)$$

proving $\text{MSE}_{\text{MMSE}} \leq \text{MSE}_{\text{LMMSE}}$.


$$\text{Cov}(x, Y) = \text{Cov}(\hat{x}_{\text{MMSE}}, Y)$$

$$= E[\bar{x} \bar{Y} - \bar{\hat{x}}_{\text{MMSE}} \bar{Y}]$$

$$= E[(\bar{x} - \bar{\hat{x}}_{\text{MMSE}}) \bar{Y}]$$

total law
of expectations

$$= E \left[E \left[(\bar{x} - \hat{x}_{\text{MMSE}}) | Y \right] \right]$$

as $E[f(x)g(y)|y] \leftarrow = E \left[\bar{y} E[\bar{x} - \hat{x}_{\text{MMSE}} | Y] \right]$

$$= g(y) E[f(x)|y] = 0.$$

$$E[\bar{x} - \hat{x}_{\text{MMSE}} | Y]$$

$$= E[\bar{x} | Y] - \hat{x}_{\text{MMSE}}$$

$$= \hat{x}_{\text{MMSE}} - \cancel{E[x]} - \hat{x}_{\text{MMSE}} + \cancel{E[\hat{x}_{\text{MMSE}}]}$$

$$= 0.$$

We have shown
something similar
earlier

$$E[x - \hat{x}_{\text{MMSE}} | Y] = 0.$$